
myDAQ

Specifications

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Definitions

Warranted specifications describe the performance of a model under stated operating conditions and are covered by the model warranty.

Characteristics describe values that are relevant to the use of the model under stated operating conditions but are not covered by the model warranty.

- *Typical* specifications describe the performance met by a majority of models.
- *Nominal* specifications describe an attribute that is based on design, conformance testing, or supplemental testing.

Specifications are *Typical* unless otherwise noted.

Conditions

Specifications are valid at 25 °C unless otherwise noted.

Analog Input

Number of channels	2 differential or 1 stereo audio input
ADC resolution	16 bits
Maximum sampling rate	200 kS/s
Timing accuracy	100 ppm of sample rate

Timing resolution		10 ns
Range		
Analog input	±10 V, ±2 V, DC-coupled	
Audio input	±2 V, AC-coupled	
Passband (-3 dB)		
Analog input	DC to 400 kHz	
Audio input	1.5 Hz to 400 kHz	
Connector type		
Analog input	Screw terminals	
Audio input	3.5 mm stereo jack	
Input type (audio input)		Line-in or microphone
Microphone excitation (audio input)		5.25 V through 10 kΩ

Table 1. Absolute accuracy

Nominal Range		Typical at 23 °C (mV)	Maximum (18 to 28 °C) (mV)
Positive Full Scale	Negative Full Scale		
10	-10	22.8	38.9
2	-2	4.9	8.6

Figure 1. Settling Time (10 V Range) versus Different Source Impedance

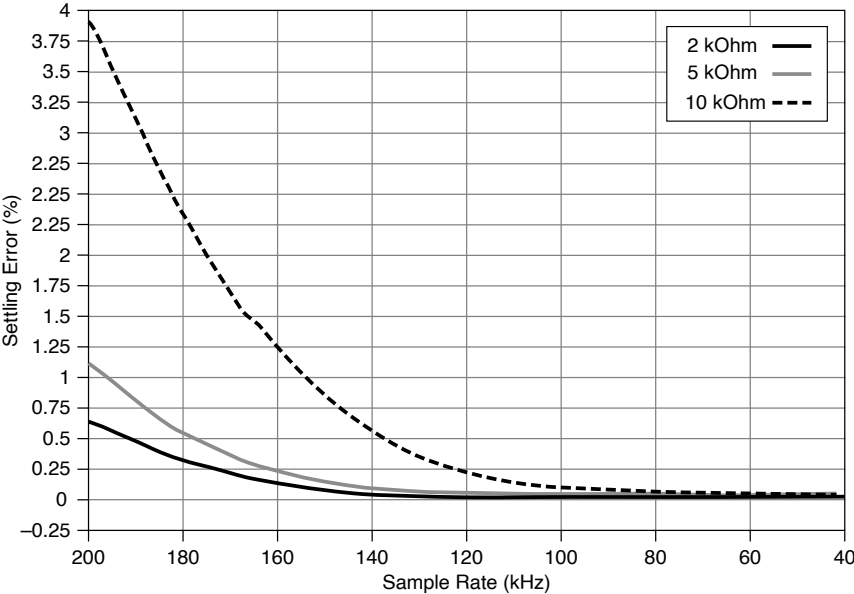
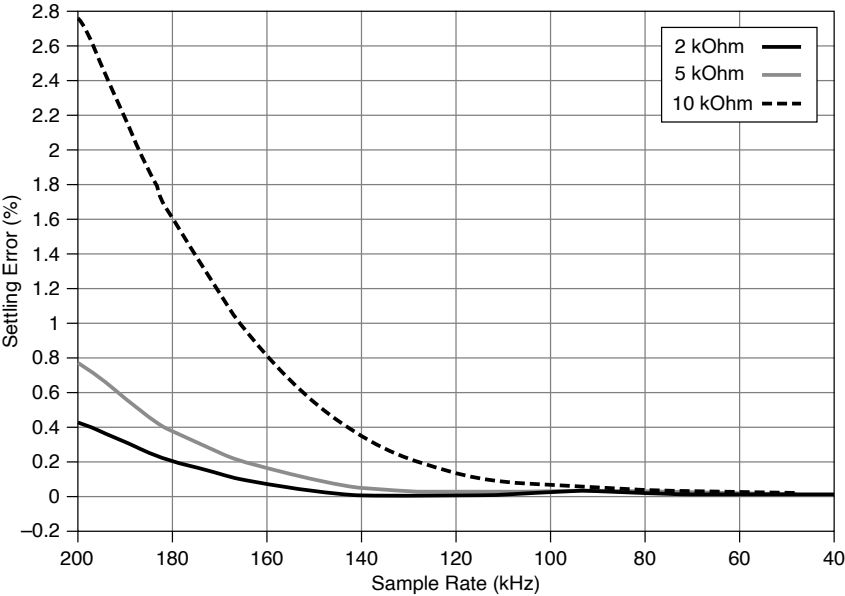


Figure 2. Settling Time (2 V Range) versus Different Source Impedance



Input FIFO size	4,095 samples, shared among channels used
Maximum working voltage for analog inputs (signal + common mode)	±10.5 V to AGND

Common-mode rejection ratio (CMRR) (DC to 60 Hz)	70 dB
Input impedance	
Device on	
AI+ or AI- to AGND	$>10\text{ G}\Omega \parallel 100\text{ pF}$
AI+ to AI-	$>10\text{ G}\Omega \parallel 100\text{ pF}$
Device off	
AI+ or AI- to AGND	5 k Ω
AI+ to AI-	10 k Ω
Anti-aliasing filter	None
Overvoltage protection	
AI+ or AI- to AGND	$\pm 16\text{ V}$
Audio input left and right	None

Analog Output

Number of channels	2 ground-referenced or 1 stereo audio output
DAC resolution	16 bits

Maximum update rate	200 kS/s
Range¹	
Analog output	± 10 V, ± 2 V, DC-coupled
Audio output	± 2 V, AC-coupled



Note Creating a task on an AO channel will set the range for both channels. If a task that was previously running is stopped and a new task is created using the second AO channel, the output on the original channel will scale based on the range of the new task.

Maximum output current (analog output) ²	2 mA
Output impedance	
Analog output	1 Ω
Audio output	120 Ω
Minimum load impedance (audio output)	8 Ω
Connector type	
Analog output	Screw terminals

1. When outputting from AO 0 and AO 1 concurrently, you must set a common voltage range for both channels.
2. The total power available for the power supplies, analog outputs, and digital outputs is limited to 500 mW (typical)/100 mW (minimum).

Audio output	3.5 mm stereo jack
AC-coupling high-pass frequency (audio output with 32 Ω load)	48 Hz

Table 2. Absolute accuracy

Nominal Range		Typical at 23 °C (mV)	Maximum (18 to 28 °C) (mV)
Positive Full Scale	Negative Full Scale		
10	-10	19.6	42.8
2	-2	5.4	8.8

Slew rate	4 V/ μ s
Timing accuracy	100 ppm of sample rate
Timing resolution	10 ns
Overdrive protection	± 16 V to AGND
Maximum power-on voltage ³	± 110 mV
Output FIFO size	8,191 samples, shared among channels used

Related reference:

- [Calculating Power Consumption](#)

3. When powered on, the analog output signal is not defined until after USB configuration is complete.

Digital I/O

Number of lines	8; DIO <0..7>
Direction control	Each line individually programmable as input or output
Update mode	Software-timed
Pull-down resistor	75 k Ω
Logic level	5 V compatible LVTTTL input; 3.3 V LVTTTL output
V _{IH} min	2.0 V
V _{IL} max	0.8 V
Maximum output current per line ⁴	4 mA

Related reference:

- [Calculating Power Consumption](#)

General Purpose Counter/Timer

Number of counter/timers	1
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4. The total power available for the power supplies, analog outputs, and digital outputs is limited to 500 mW (typical)/100 mW (minimum).

Resolution	32 bits
Internal base clocks	100 MHz
Base clock accuracy	100 ppm
Maximum counting and pulse generation update rate	1 MS/s
Default routing	
CTR 0 SOURCE	PFI 0 routed through DIO 0
CTR 0 GATE	PFI 1 routed through DIO 1
CTR 0 AUX	PFI 2 routed through DIO 2
CTR 0 OUT	PFI 3 routed through DIO 3
FREQ OUT	PFI 4 routed through DIO 4
Data transfers	Programmed I/O
Update mode	Software-timed

Digital Multimeter

Functions ⁵	DC voltage, AC voltage, DC current, AC current, resistance, diode, continuity
Isolation level	60 VDC/20 V _{rms} , Measurement Category I
Connectivity	Banana jacks
Resolution	3.5 digits
Input coupling	DC (DC Voltage, DC Current, Resistance, Diode, Continuity); AC (AC Voltage, AC Current)

Related reference:

- [Safety Voltages](#)

Voltage Measurement

DC ranges	200 mV, 2 V, 20 V, 60 V
AC ranges	200 mV _{rms} , 2 V _{rms} , 20 V _{rms}



Note All AC voltage accuracy specifications apply to signal amplitudes greater than 5% of range.

5. All AC specifications are based on sine wave RMS.

Table 3. Accuracy, DC Voltage Function

Range	Resolution	Accuracy
		$\pm ([\% \text{ of Reading}] + \text{Offset})$
200.0 mV	0.1 mV	0.5% + 0.2 mV
2.000 V	0.001 V	0.5% + 2 mV
20.00 V	0.01 V	0.5% + 20 mV
60.0 V	0.1 V	0.5% + 200 mV

Table 4. Accuracy, AC Voltage Function

Range	Resolution	Accuracy	
		$\pm ([\% \text{ of Reading}] + \text{Offset})$	
		40 to 400 Hz	400 to 2,000 Hz
200.0 mV	0.1 mV	1.4% + 0.6 mV ⁶	-
2.000 V	0.001 V	1.4% + 0.005 V	5.4% + 0.005 V
20.00 V	0.01 V	1.5% + 0.05 V	5.5% + 0.05 V

Input impedance	10 M Ω
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Current Measurement

DC ranges	20 mA, 200 mA, 1 A
AC ranges	20 mA _{rms} , 200 mA _{rms} , 1 A _{rms}

6. The accuracy for AC Volts 200.0 mV range is in the frequency range of 40 Hz to 100 Hz. For example, for a 10 V using the DC Volts function in the 20.00 V range, calculate the accuracy using the following equation: $10 \text{ V} \times 0.5\% + 20 \text{ mV} = 0.07 \text{ V}$



Note All AC accuracy specifications within 20 mA and 200 mA ranges apply to signal amplitudes greater than 5% of range. All AC accuracy specifications within the 1 A range apply to signal amplitudes greater than 10% of range.

Table 5. Accuracy, DC Current Function

Range	Resolution	Accuracy
		$\pm$ ([% of Reading] + Offset)
20.00 mA	0.01 mA	0.5% + 0.03 mA
200.0 mA	0.1 mA	0.5% + 0.3 mA
1.000 A	0.001 A	0.5% + 3 mA

Table 6. Accuracy, AC Current Function

Range	Resolution	Accuracy	
		$\pm$ ([% of Reading] + Offset)	
		40 to 400 Hz	400 to 2,000 Hz
20.00 mA	0.01 mA	1.4% + 0.06 mA	5% + 0.06 mA
200.0 mA	0.1 mA	1.5% + 0.8 mA	5% + 0.8 mA
1.000 A	0.001 A	1.6% + 6 mA	5% + 6 mA

Input protection	Internal ceramic fuse, 1.25 A 250 V, fast-acting, 5 × 20 mm, F 1.25A H 250V (Littelfuse part number 02161.25)
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Resistance Measurement

Ranges	200 Ω , 2 k Ω , 20 k Ω , 200 k Ω , 2 M Ω , 20 M Ω
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Table 7. Accuracy, Resistance (Ω) Function

Range	Resolution	Accuracy
		$\pm$ ([% of Reading] + Offset)
200.0 Ω	0.1 Ω	0.8% + 0.3 Ω ⁷
2.000 k Ω	0.001 k Ω	0.8% + 3 Ω
20.00 k Ω	0.01 k Ω	0.8% + 30 Ω
200.0 k Ω	0.1 k Ω	0.8% + 300 Ω
2.000 M Ω	0.001 M Ω	0.8% + 3 k Ω
20.00 M Ω	0.01 M Ω	1.5% + 50 k Ω

Diode Measurement

Range	2 V
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Power Supplies



Caution Do not mix power from NI myDAQ with power from external power sources. When using external power, remove any connections to the power supply terminals on NI myDAQ.

Related reference:

- [Calculating Power Consumption](#)

+15V Supply

Output voltage	
Typical (no load)	15.0 V

7. Exclusive of lead wire resistance

Maximum voltage (no load)	15.3 V
Minimum voltage (full load)	14.0
Maximum output current ⁸	32 mA
Maximum load capacitance	470 μ F

-15V Supply

Output voltage	
Typical (no load)	-15.0 V
Maximum voltage (no load)	-15.3 V
Minimum voltage (full load)	-14.0
Maximum output current ⁹	32 mA
Maximum load capacitance	470 μ F

8. The total power available for the power supplies, analog outputs, and digital outputs is limited to 500 mW (typical)/100 mW (minimum).
9. The total power available for the power supplies, analog outputs, and digital outputs is limited to 500 mW (typical)/100 mW (minimum).

+5V Supply

Output voltage	
Typical (no load)	4.9 V
Maximum voltage (no load)	5.2 V
Minimum voltage (full load)	4.0
Maximum output current ¹⁰	100 mA
Maximum load capacitance	33 μ F

Calculating Power Consumption

The total power available for the power supplies, analog outputs, and digital outputs is limited to 500 mW (typical)/100 mW (minimum). To calculate the total power consumption of the power supplies, multiply the output voltage by the load current for each voltage rail and sum them together. For digital output power consumption, multiply 3.3 V by the load current. For analog output power consumption, multiply 15 V by the load current. Using audio output subtracts 100 mW from the total power budget.

If you use 50 mA on +5 V, 2 mA on +15 V, 1 mA on -15 V, use four DIO lines to drive LEDs at 3 mA each, and have a 1 mA load on each AO channel, the *total output power consumption* is 364.6 mW

- $5\text{ V} \times 50\text{ mA} = 250\text{ mW}$
- $|+15\text{ V}| \times 2\text{ mA} = 30\text{ mW}$
- $|-15\text{ V}| \times 1\text{ mA} = 15\text{ mW}$

10. The total power available for the power supplies, analog outputs, and digital outputs is limited to 500 mW (typical)/100 mW (minimum).

- $3.3\text{ V} \times 3\text{ mA} \times 4 = 39.6\text{ mW}$
- $15\text{ V} \times 1\text{ mA} \times 2 = 30\text{ mW}$
- *Total output power consumption* = $250\text{ mW} + 30\text{ mW} + 15\text{ mW} + 39.6\text{ mW} + 30\text{ mW} = 364.6\text{ mW}$

Related reference:

- [Analog Output](#)
- [Digital I/O](#)
- [Power Supplies](#)

Communication

Bus interface	USB 2.0 Hi-Speed
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Physical Characteristics

Clean the hardware with a soft, nonmetallic brush. Make sure that the hardware is completely dry and free from contaminants before returning it to service.

Dimensions (without screw terminal connector)	
NI myDAQ device part number 195509D-01L and earlier	14.6 cm × 8.7 cm × 2.2 cm (5.75 in. × 3.43 in. × 0.87 in.)
NI myDAQ device part number 195509E-01L and later	13.6 cm × 8.8 cm × 2.4 cm (5.36 in. × 3.48 in. × 0.95 in.)
Weight	
NI myDAQ device part number 195509D-01L and earlier	175.0 g (6.1 oz)
NI myDAQ device part number 195509E-01L and later	164.0 g (5.8 oz)



Note NI myDAQ device part number (P/N: 195509x-01L) is located on the product label on the bottom of the device.

Screw-terminal wiring	16 to 26 AWG
Torque for screw terminals	0.22-0.25 N · m (2.0-2.2 lb · in.)

Environmental

Temperature (IEC 60068-2-1 and IEC 60068-2-2)	
Operating	0 to 45 °C
Storage	-20 to 70 °C
Humidity (IEC 60068-2-56)	
Operating	10 to 90% RH, noncondensing
Storage	10 to 90% RH, noncondensing
Pollution Degree (IEC 60664)	2
Maximum altitude	2,000 m (at 25 °C ambient temperature)

Indoor use only.

Safety Voltages

Related reference:

- [Digital Multimeter](#)

Safety Compliance Standards

This product is designed to meet the requirements of the following electrical equipment safety standards for measurement, control, and laboratory use:

- IEC 61010-1, EN 61010-1
- UL 61010-1, CSA C22.2 No. 61010-1



Note For safety certifications, refer to the product label or the [Product Certifications and Declarations](#) section.

Hazardous Locations

This device is not certified for use in hazardous locations.

Electromagnetic Compatibility

This product meets the requirements of the following EMC standards for electrical equipment for measurement, control, and laboratory use:

- EN 61326-1 (IEC 61326-1): Class A emissions; Basic immunity
- EN 55011 (CISPR 11): Group 1, Class A emissions
- AS/NZS CISPR 11: Group 1, Class A emissions
- FCC 47 CFR Part 15B: Class A emissions
- ICES-001: Class A emissions



Note In the United States (per FCC 47 CFR), Class A equipment is intended for use in commercial, light-industrial, and heavy-industrial locations. In Europe, Canada, Australia and New Zealand (per CISPR 11) Class A equipment is intended for use only in heavy-industrial locations.



Note Group 1 equipment (per CISPR 11) is any industrial, scientific, or

medical equipment that does not intentionally generate radio frequency energy for the treatment of material or inspection/analysis purposes.



Note For EMC declarations and certifications, and additional information, refer to the [Online Product Certification](#) section.

Product Certifications and Declarations

Refer to the product Declaration of Conformity (DoC) for additional regulatory compliance information. To obtain product certifications and the DoC for NI products, visit ni.com/product-certifications, search by model number, and click the appropriate link.

Environmental Management

NI is committed to designing and manufacturing products in an environmentally responsible manner. NI recognizes that eliminating certain hazardous substances from our products is beneficial to the environment and to NI customers.

For additional environmental information, refer to the *Engineering a Healthy Planet* web page at ni.com/environment. This page contains the environmental regulations and directives with which NI complies, as well as other environmental information not included in this document.

Waste Electrical and Electronic Equipment (WEEE)



EU Customers This symbol indicates that waste products should be disposed of separately from municipal household waste according to WEEE Directive 2002/96/EC of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources. For information about the available collection and

recycling scheme in a particular country, go to ni.com/environment/weee.

电子信息产品污染控制管理办法（中国RoHS）

-  中国RoHS—NI符合中国电子信息产品中限制使用某些有害物质指令 (RoHS)。关于NI中国RoHS合规性信息，请登录 ni.com/environment/rohs_china。 (For information about China RoHS compliance, go to ni.com/environment/rohs_china.)